Alekya Sai Laxmi Kowta

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Software Engineer — Distributed Systems — Cloud & Microservices

SUMMARY

Software Engineer specializing in fault-tolerant distributed systems, cloud-native microservices, and event-driven data pipelines using Kubernetes, Kafka, and AWS/Azure. Experienced architecting scalable, high-availability backend infrastructure for global traffic.

EDUCATION

The George Washington University *Expected May 2027*
M.S. in Computer Science (GPA: 3.9/4.0) — Washington, DC

Vellore Institute of Technology (VIT) *May 2022*
B.Tech in Information Technology — Tamil Nadu, India

TECHNICAL SKILLS

Languages: Java, C#, .NET, Python, C++, JavaScript, SQL
Frameworks & Web: Spring Boot, ASP.NET Core MVC, REST APIs, Microservices, HTML, CSS
Databases: Oracle, MySQL, NoSQL, Redis
Tools & DevOps: Git, Jenkins, Docker, Kubernetes, PowerBI, Grafana, Jira, Confluence, Linux
Cloud & Messaging: AWS (EC2, Lambda, SQS, ElastiCache, API Gateway, S3), Microsoft Azure, Kafka, RabbitMQ, Azure Event Hubs
AI & Data: OpenAI GPT-4o, NVIDIA NIM, Multi-Agent Systems, Pandas, AI Data Analysis
Methodologies: Agile/Scrum, TDD, Unit & Integration Testing, Code Reviews, High-Throughput Distributed Systems

WORK EXPERIENCE

Software Engineer I, Dell Technologies — *Bangalore, India* ***Aug 2022 – Aug 2025***

- Architected distributed REST APIs to remotely configure 150M+ connected devices, ensuring 99.9% availability for global traffic
- Led infrastructure migration to Kubernetes, optimizing CI/CD pipelines to achieve 20% throughput gain and 35% reduction in deployment latency across microservices
- Built high-throughput Kafka and Azure Event Hubs pipelines processing 500K+ real-time telemetry events per day with sub-second latency
- Reduced data retrieval latency by 40% implementing Redis caching layer for high-concurrency key-value storage across distributed microservices
- Designed Oracle/SQL and NoSQL schemas supporting 10M+ daily transactions with zero downtime during peak load

Software Engineering Intern, Dell Technologies — *Bangalore, India* ***Jan 2022 – Aug 2022***

- Built real-time Grafana and PowerBI dashboards monitoring 15+ service KPIs, reducing incident detection time by 25%
- Automated deployment validation using Python scripts and co-developed 3 cross-team backend POCs for secure workspace features, saving ∼4 hours per release cycle

Software Intern, GKMT IT — *Remote* ***Mar 2020 – Jun 2020***

- Engineered responsive web platforms reducing page load times by 25%; designed secure authentication system cutting account-related support tickets by 40%

PROJECTS

Debatrium — Distributed Multi-Agent AI Debate System *Python, AWS, OpenAI GPT-4o, NVIDIA NIM, SQS, Redis*

- Architected a 9-agent cloud-native system on AWS across three tiers (Research, Critic, Judge) using dedicated EC2 Auto Scaling Groups, 12 SQS FIFO queues with DLQs, and ElastiCache Redis 7.0 (Multi-AZ, TLS) for fault-tolerant state management
- Engineered iterative quality-control loop where Judge Agents score debates on fairness, accuracy, and consensus; scores below 0.85 automatically re-inject feedback into Research Agents for up to 3 refinement rounds, framing answer quality as an optimization problem

Multi-Objective Carbon-Aware Freight Routing *Python, Pandas, OSMnx, PyTorch, Stable-Baselines3*

- Built an end-to-end pipeline enriching a real DC road graph (∼960 nodes) with MOVES5 emission factors, SRTM elevation grades, and 4 time-of-day congestion profiles, generating 12 carbon weight attributes per edge across 3 vehicle classes
- Benchmarked Weighted A* (3.34% carbon savings, 100% routing success), Double DQN (120K training steps), and PPO with Behavioral Cloning (0.946 carbon optimality) jointly optimizing delivery time and CO_2 via preference weight $\alpha \in [0, 1]$; terrain study showed 85% more benefit in hilly vs. flat networks

Trustworthy AI Hackathon — 1st Place Winner *Python, Pandas, Jupyter* | **Link**

- Awarded 1st Prize for an open-source analytical framework democratizing access to the AI Incident Database; engineered reproducible Jupyter notebooks and Pandas visualizations translating complex AI-harm datasets into summary statistics for non-technical stakeholders

PUBLICATIONS & PATENTS

- Kowta, A. S. L. (2022). Analysis and overview of information gathering & tools for pentesting. *IEEE Conference* | **Link**
- Kowta, A. S. L. (2022). Cyber security and the internet of things: Vulnerabilities, threats, intruders, and attacks. *Cyber Intelligence and Information Retrieval*, Springer | **Link**
- Kowta, A. S. L. (2022). Smart walking system for the elderly and the blind. *Patent Application*